- What is the problem you want to solve?
  - Current robot manipulation tasks, demonstrations and research heavily relies on operator to hand over the object into the hand of the robot manipulator in order for the manipulator to carry out experiments and trajectories.
  - I want to solve the problem which robot manipulators having trouble picking up soft and thin objects such as cloths.
- Why should engineers and roboticists care about this problem?
  - If robot graspers are not able to pick up cloths flat from the surface, the "automonous" nature of robots will be highly limited, manipulations methods will be limited.
  - The ability to grasp cloth-like objects enables:
    - Laundry folding and textile handling
    - Hospital bedding management
    - Assistive robotics for elderly or disabled individuals
    - Garment manufacturing and logistics
    - Soft object packaging
- What is the current, state-of-the-art solution to the problem?
  - Current state-of-the-art solutions combine environment interaction, specialized end-effectors, and feedback mechanisms to enable reliable cloth grasping.
  - Environment-assisted actions are used to create graspable features (e.g., folds, wrinkles, or edges) through dragging, wiping, or peeling, allowing standard grippers to perform pinch grasps.
  - Specialized end-effectors reduce reliance on precise edge access, including variable-friction grippers, micro-needle or needle-based graspers, electroadhesion, and airflow-based methods (e.g., Bernoulli or Coandă effects).
  - Closed-loop perception and control are often integrated to verify grasp success (e.g., single-ply detection) and trigger recovery actions when failures occur.
  - In practice, most systems still rely on rigid parallel-jaw grippers augmented with these strategies.
  - Some work existed working on comparison between the performance of soft and parallel jaw grippers, however, the contact part were either not addressed or require active control on the soft grasper.
- What are the drawbacks of the state-of-the-art?
  - Heavy reliance on environment-assisted actions reduces true autonomy and increases task complexity and execution time.
  - Specialized end-effectors are often task-specific, difficult to generalize, and may be impractical or fragile for real-world deployment.
  - Rigid parallel-jaw grippers require precise positioning and well-defined geometry, making them unreliable for thin, flat cloth.

- Closed-loop verification and recovery introduce additional sensing and control complexity, increasing system overhead.
- Most approaches treat the gripper as fixed and focus on perception or planning, resulting in a lack of systematic evaluation of how gripper design (e.g., rigid vs soft) affects performance.
- Consequently, directly grasping thin, flat cloth from a surface remains an open and unresolved problem.

- What is your contribution?
  - I wish to create a benchmark focusing on the “contact” part of the grasping of soft material, comparing the performance between soft but adaptive (passive, which did not add more control space compare to parallel jaw) and the parallel jaw gripper.
    - The benchmark will include: (“control policy agrees” means that under the lab franka settings, the controller did not “quit” due to error)
      - Offset from the table that the control policy may operate and there success rate of grasping the cloth – z axis tolerance
      - Angular tolerance (how much the robot grasper head could rotate before policy fails)
      - Force that could be exerted to the surface while control policy may agrees
      - Peak force during successful grasps (that were exerted to table or cloth)
      - Minimum contact force required for a grasp (that were exerted to table or cloth)
    - Apart from the benchmark, I want to test both graspers on the same human-defined policy for specific tasks:
      - Fold cloth diagonally (flat cloth corner grasp)
      - Grasp the cloth from middle and drop it onto a sphere next to it (flat cloth middle grasp)
      - Drag the cloth from position A to position B (flat cloth edge grasp)
        - Some more difficult tasks may include dealing with the folded cloth.
- How will your contribution address the drawbacks?
  - By introducing a benchmark that explicitly treats the gripper as a variable, this work addresses the current limitation where hardware is assumed fixed and not systematically evaluated.
  - The proposed metrics (e.g., positional tolerance, angular tolerance, and force profiles) directly quantify how different grippers behave under uncertainty, reducing reliance on environment-assisted actions such as dragging or folding to create graspable features.
  - By comparing a soft, passive adaptive gripper with a rigid parallel-jaw gripper under the same control policy, the study isolates the effect of compliance and contact mechanics, rather than conflating it with perception or planning improvements.

- The inclusion of force-related metrics (minimum, peak, and allowable forces) provides insight into failure modes such as slipping, snagging, or excessive force, which are not captured in current benchmarks.
- The task-based evaluation ensures that grasps are not only successful at contact, but also useful for downstream manipulation, addressing the gap between grasp acquisition and task execution.
- Overall, this contribution shifts part of the problem from complex perception and control toward hardware-aware design, potentially reducing system complexity while improving robustness in manipulating thin, deformable objects such as cloth.

Strategy of grasping:

1. Multistage descent: Assuming the contact point is at height 0, first descent fast to the x,y position of the contact, with height 5mm, slowly descent to height 0 mm, then slower descent to a small amount of negative height (e.g. 3mm), in a very slow fashion, this would allow safety margin of contact as well as force limit, the grasper will also deform slightly in this stage, and the contact force between the contact points of the grasper and cloth will allow next few steps to be possible
2. Slowly close the grasper, as there is contact force between the lower section of the grasper and the cloth, this closing movement would allow grasper to "pinch" the cloth into the inner contact surface of the grasper.
3. Slowly ascent the grasper in the first few steps as this would allow time for grasper to recover to its original shape and grab onto the cloth firmly.
4. In these steps, the margin of grasper opening distance would mean the amount of fabric being pinched and sent into grasper for contact.
5. All these steps are made possible by the deformable nature of finray. When using original parallel jaw grippers, it also succeeded, but do not require the extra step of "moving into the negative height", a more careful tunning of height parameter is needed, before the execution fails due to excessive contact force or too lose of contact force resulting in failure of grasping.